

SCANPOS

Product Requirements Document

Supermarket Point-of-Sale & Inventory Management App

Version	1.0 — Initial Draft
Date	May 2026
Status	Ready for Development
Platform	Mobile (iOS & Android)

1. Product Overview

ScanPos is a mobile point-of-sale (POS) and inventory management app designed for supermarkets. It allows cashiers to scan product barcodes or QR codes, calculate totals, and process sales — while giving managers and warehouse staff full control over the product catalog and stock levels.

1.1 Core Value Proposition

- Fast, frictionless scanning-based checkout experience
- Real-time inventory tracking — stock auto-decrements on every sale
- Role-based access for managers, warehouse staff, and cashiers
- Clean black-and-white UI with rounded, modern design language

1.2 Design Principles

- Color palette: Pure black & white only — no color accents
- Shape language: Rounded corners, circular elements, soft curves throughout
- Minimalist navigation: Bottom bar with exactly two tabs — Scanner and Inventory
- Mobile-first: Designed for phones held in one hand at a checkout counter

2. User Roles & Permissions

There are three roles in ScanPos. Every user signs in with Google, then is classified into one of these roles.

Permission	Manager	Warehouse Staff	Cashier
Scan & sell products	✓	✓	✓
View sales history	✓	✗	✗
Manage inventory	✓	✓	✗
Add/edit products	✓	✓	✗
Adjust stock quantities	✓	✓	✗
Generate invite codes	✓	✗	✗
Set employee permissions	✓	✗	✗

2.1 Manager

- Creates the store account via Google Sign-In
- Gets a unique store invite code to share with employees
- Assigns each employee a role (Cashier or Warehouse Staff) after they join
- Has full access to all features including sales history

2.2 Warehouse Staff

- Signs in with Google, selects 'Employee', enters the manager's invite code
- Can browse, add, edit, and update all product data and stock quantities
- Cannot view sales history or manage other employees

2.3 Cashier

- Signs in with Google, selects 'Employee', enters the manager's invite code
- Can only scan products and process sales
- Cannot view or modify inventory or sales history

3. Authentication Flow

3.1 First-Time Sign-In

Step 1 — Google Sign-In: User taps 'Sign in with Google' and authenticates.

Step 2 — Role Selection: User is presented with two options:

- I'm a Manager — Create a new store
- I'm an Employee — Join an existing store

Step 3a (Manager): Store profile is created. A unique invite code is generated and shown to the manager.

Step 3b (Employee): User enters the invite code. The system matches them to the store. Manager assigns their role (Cashier or Warehouse Staff).

3.2 Returning Users

- Google Sign-In automatically restores the user's role and store context
- No re-entry of invite codes required

4. Screens & Features

4.1 Scanner Screen (QR Tab)

This is the default home screen for all users. It opens the device camera and is ready to scan immediately on launch.

Scanning Behavior

- Camera opens automatically when the tab is active
- Scans standard barcodes (EAN-13, EAN-8, UPC-A) and QR codes printed on products
- On successful scan, the product is looked up from the store's inventory by barcode
- If found: product name, price, and image are added to the current session cart
- If not found: a clear error message is shown — 'Product not in inventory'

Cart & Total

- Each scanned product appears in a running list below the scanner
- The total amount updates in real time as items are added
- Each item in the list shows: product name, unit price, quantity scanned
- User can remove an item from the cart by swiping or tapping a remove button

Pay Button

- A prominent 'Pay' button is always visible at the bottom of the screen
- Tapping 'Pay' confirms the sale
- The transaction is saved to sales history with: date, time, list of items, and total
- Stock is automatically decremented for each sold product
- The cart is cleared and the scanner returns to idle — ready for the next customer

4.2 Inventory Screen (Warehouse Tab)

Accessible by Manager and Warehouse Staff. Cashiers do not see this tab.

Product List

- Displays all products in the store, filterable by category
- Each product card shows: thumbnail image, name, category, price, stock quantity
- Search bar to find products by name or barcode number

Add Product

- Floating action button opens 'Add Product' form
- Fields: Product Name (required), Price (required), Quantity in Stock (required), Barcode Number (required), Category (required), Product Image (optional — from camera or gallery)
- Barcode field can be filled by typing or by scanning with the camera

Edit Product

- Tapping any product opens an edit sheet with all fields pre-filled
- Manager and Warehouse Staff can update: name, price, quantity, image, category
- Deleting a product removes it from inventory (with confirmation dialog)

4.3 Sales History (Manager Only)

Accessed via a dedicated section inside the Manager's profile or settings — not a bottom tab.

- Lists all completed transactions in reverse chronological order
- Each transaction shows: transaction ID, date & time, cashier name, number of items, total amount
- Tapping a transaction expands it to show the full item breakdown
- Manager can scroll and browse — no deletion or editing of history

4.4 Settings & Team Management (Manager Only)

- View all employees linked to the store
- Change any employee's role (Cashier ↔ Warehouse Staff)
- Regenerate or copy the store invite code
- Remove an employee from the store

5. Product Data Model

Field	Type	Notes
Product Name	String	Required. Displayed on scanner and inventory screens.
Price	Number (float)	Required. Price per unit in local currency.
Quantity in Stock	Integer	Required. Decrements automatically on each sale.
Barcode Number	String	Required. Must be unique per store. Used for QR/barcode lookup.
Category	String	Required. Used for filtering in inventory view.
Product Image	Image URL	Optional. Uploaded from camera or gallery.

6. Feature Summary Table

Feature	Description	Access
Google Sign-In	OAuth 2.0 authentication, no passwords	All users
Role Selection	Manager or Employee (with invite code)	All users — first login only
Invite Code System	Manager generates code, employees enter it	Manager (generate), Staff (enter)
Role Assignment	Manager assigns Cashier or Warehouse Staff	Manager only
Barcode/QR Scanning	Camera scans standard barcodes from products	All users
Running Cart	Real-time list of scanned items with total	All users
Remove Cart Item	Swipe or tap to remove before paying	All users
Pay & Confirm Sale	Saves transaction, decrements stock, clears cart	All users
Product Catalog	Browse all products with search & category filter	Manager, Warehouse Staff
Add Product	Form with name, price, qty, barcode, category, image	Manager, Warehouse Staff
Edit Product	Update any field on existing product	Manager, Warehouse Staff
Delete Product	Remove product with confirmation	Manager, Warehouse Staff
Auto Stock Deduction	Quantity decrements on every sale	Automatic
Sales History	View all past transactions with item breakdown	Manager only
Employee Management	View team, change roles, remove employees	Manager only

7. Scope

7.1 In Scope (V1)

- Google Sign-In authentication
- Three-role system: Manager, Warehouse Staff, Cashier
- Barcode and QR code scanning using device camera
- Real-time cart with total and pay flow
- Full inventory management (add, edit, delete products)
- Auto stock deduction on sale
- Sales history for managers
- Invite code onboarding for employees
- Black & white rounded UI

7.2 Out of Scope (Future Versions)

- Payment gateway integration (cash-only in V1)
- Receipt printing or PDF export
- Barcode/QR code generation for custom labels
- Analytics dashboard and sales reports
- Multi-branch / multi-store support
- Low stock alerts and notifications
- Customer loyalty or discount system

8. Suggested Tech Stack

Frontend	React Native (cross-platform iOS & Android)
Authentication	Google OAuth via Firebase Auth
Database	Firebase Firestore (real-time, per-store collections)
Storage	Firebase Storage (product images)
Barcode Scanning	react-native-vision-camera + Vision Camera Barcode Plugin
State Management	Zustand or Redux Toolkit
Navigation	React Navigation v6 (bottom tabs + stack)